ANN-KATHRIN SCHUETZ

PhD in Physics | Research Scientist, Physical AI | Simulation, Trustworthy ML & Real-World Sensors

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I build ML that makes expensive physical simulation fast and reliable enough to drive real engineering decisions, and I test it on hardware I've built myself. PhD in physics, five years at Berkeley Lab, UC Berkeley master's in data science. I work across the whole stack, from sensors and electronics to models and GPU clusters.

HIGHLIGHTS

- Built an ML-driven optimization method for rare-event simulations; it found a detector design with **66.5% less noise** using **30× less compute**. First-author **ICLR 2025 Spotlight**; adopted by multiple research collaborations.
- Showed the method transfers to a different scientific domain: on extremely imbalanced data (under **0.01%** positives), it cut compute by **98%** and enabled previously infeasible high-dimensional parameter inference.
- Took sensor hardware from design to deployment: electronics, cleanroom fabrication, commissioning. **Berkeley Lab SPOT Award**.

EXPERIENCE

Lawrence Berkeley National Laboratory May 2021 – Present
Postdoctoral Researcher, Research Scientist

- **Made rare-event design optimization affordable.** Built RESuM, an open-source ML surrogate that learns intractable event probabilities directly from simulation and combines physics models of different fidelity to give calibrated, uncertainty-aware predictions. Used it to optimize detector shielding in the search for a nuclear decay with a half-life longer than the age of the universe, cutting rare noise from cosmic-ray showers by **66.5%** at **3.3% of standard compute**. Adopted by multiple scientific collaborations. Paper | Code | Video
- **Replaced a slow simulator with a fast ML stand-in.** Repurposed RESuM as a likelihood emulator for binary black-hole merger outcomes rarer than 1 in 10,000, showing it works beyond design optimization. It cut compute by **98%** and enabled the first full high-dimensional parameter inference, making population-synthesis studies of stellar evolution feasible.
- **Automated the step from engineering documents to runnable simulation.** Built an LLM agent that turns sparse, incomplete multimodal detector documentation into physically consistent 3D geometries and executable simulations, combining knowledge graphs, hierarchical spatial representations, and automated geometric validation.
- **Brought ML toward real-time sensor decisions.** Developed and benchmarked deep-learning models that classify waveforms (time-series signals) from germanium semiconductor sensors to reject noise, and optimized their inference latency for integration into the data acquisition system.
- **Owned hardware end to end.** Led the design, cleanroom fabrication (thin-film sputtering, photolithography, trace etching), testing, installation, and commissioning of ultra-low-noise electronics and data acquisition systems.
- **Built reproducible research software and scaled it on supercomputers.** Wrote research code in Python and C++ (PyTorch, TensorFlow) and ran distributed simulation and training pipelines at NERSC, the U.S. Department of Energy's national supercomputing center, using Slurm, CUDA, and OpenMP.
- **Led and mentored.** Early Career Representative for the international LEGEND collaboration (2021–2023). Mentored students and presented at international conferences and ML workshops.

University of Tübingen, Germany Jan 2014 – May 2021

Postdoctoral Researcher (2019–2021) · PhD Researcher (2014–2019)

- **Simulated semiconductor sensor physics.** Contributed to electric-field and charge-drift simulations of how signals form in semiconductor sensors; developed signal-processing algorithms to separate signal from noise.
- **Pioneered a multivariate Bayesian noise model.** Combined heterogeneous datasets and systematic uncertainties into a unified model quantifying each noise source with full uncertainty estimates.
- **Developed large-scale Monte Carlo simulations.** Built modular C++ frameworks for particle transport, radioactive decays, and cosmic-ray interactions in complex 3D geometries, validated against measured data.
- **Built a real-time sensor data pipeline.** Operated a photosensor data acquisition system and developed its full data flow, from raw signals through digital signal processing to real-time noise flagging.

ROBOTICS & APPLIED ML PROJECTS

- **Animal2robot: teaching a Unitree Go2 to walk like a real dog** (independent, early stage). Building a pipeline to extract gait patterns from dog video and train RL locomotion policies in Isaac Lab; working with ROS 2 on my own Go2 toward deploying them on the robot. Code
- **OmniSearch: robot and drone teams for wildfire search-and-rescue** (master's project). Built a dynamic simulator and multi-agent RL so robots and drones coordinate search without a central controller; trained a YOLO detector to find human survivors in aerial imagery. Code

SELECTED PUBLICATIONS

A.-K. Schuetz, et al., *RESuM: Rare Event Surrogate Model for Physics Detector Design.* ICLR 2025 (**Spotlight**).

H. Kim, S. Mak, **A.-K. Schuetz**, A. W. P. Poon, *Efficient optimization of expensive black-box simulators via marginal means, with application to neutrino detector design*, arXiv:2508.01834, August 2025.

A. Drobizhev, …, **A.-K. Schuetz**, et al., *Characterization of the First Prototype of the L1K65n Differential-Output Charge-Sensitive Preamplifier ASIC for … HPGe Detector Readout*, IEEE Trans. Nucl. Sci. 73 (2026).

EDUCATION

University of California, Berkeley · Master of Information and Data Science · GPA 3.91 2026

Coursework: Computer Vision, Applied Machine Learning, Generative AI, Neurosymbolic AI, ML at Scale

University of Tübingen (Germany) · PhD in Physics 2019

Degree awarded magna cum laude (with great distinction)

University of Tübingen (Germany) · BSc and MSc in Physics 2013

SKILLS

Agentic AI & LLMs: multi-agent systems, LangGraph, LangChain, RAG, LoRA/QLoRA, knowledge graphs.
Deep learning: PyTorch, TensorFlow/Keras, scikit-learn, transformers, diffusion models, VAEs, CNNs, multimodal representation learning.
RL & robotics: PPO, HAPPO, multi-agent RL, decentralized coordination, path planning, Isaac Lab, Isaac Sim, VMAS, ROS/ROS 2.
Perception & signal processing: YOLO, point clouds, stereo geometry, homography, motion estimation; DSP, waveform analysis, noise filtering, time- and frequency-domain analysis, streaming DAQ, low-latency inference.
Probabilistic ML & simulation: GPy, Gaussian processes, Bayesian inference, MCMC, uncertainty quantification, surrogate models, Monte Carlo, physics-based 3D simulation (Geant4).
Hardware: ultra-low-noise electronics, cleanroom fabrication, sensor characterization, sensor-response modeling, installation and commissioning.
Languages: Python, C++, Julia, R, SQL, Bash.
Infrastructure & MLOps: Docker, Singularity, Kubernetes, AWS, Spark, Airflow, Snakemake, MongoDB, Git, Linux.
HPC & GPU: CUDA, Slurm, OpenMP, distributed training on DOE supercomputers (NERSC).